Graph Theory Notes

Jesse Oldroyd

July 2, 2026

Graph Theory Notes

Jesse Oldroyd
July 2, 2026

Outline

Outline

Introduction to Graph Theory

Trees

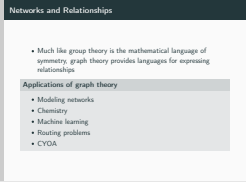
Introduction to Graph Theory

Networks and Relationships

- Much like group theory is the mathematical language of symmetry, graph theory provides languages for expressing relationships

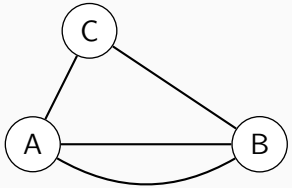
Applications of graph theory

- Modeling networks
- Chemistry
- Machine learning
- Routing problems
- CYOA

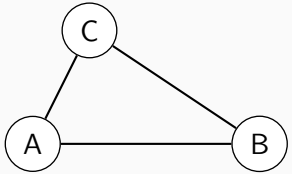


Terminology

- A graph $\mathcal{G} = (V, E)$ is a set V of vertices together with a set E of edges. Two vertices are adjacent or neighbors if they are connected by an edge.
- A simple graph is a graph with at most one edge between any pair of vertices and no loops.



(a) A graph that is not simple.



(b) A graph that is simple.

Terminology
• A graph $\mathcal{G} = (V, E)$ is a set V of vertices together with a set E of edges. Two vertices are adjacent or neighbors if they are connected by an edge.
• A simple graph is a graph with at most one edge between any pair of vertices and no loops.
(a) A graph that is not simple. (b) A graph that is simple.

Friendship

- A friendship graph is a graph where each pair of vertices have exactly one neighbor (i.e., the "friend") in common.

Example (Friendship graphs)

Draw examples of friendship graphs.

Friendship
• A friendship graph is a graph where each pair of vertices have exactly one neighbor (i.e., the "friend") in common.
Example (Friendship graphs)
Draw examples of friendship graphs.

Degrees, Paths and Cycles

- The degree of a vertex is the number of edges adjacent to that vertex. A path between two vertices is a series of distinct, adjacent vertices that connects the vertices. A cycle is a path that starts and ends at the same vertex.

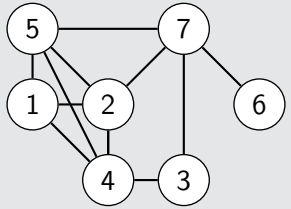
Degrees, Paths and Cycles
• The degree of a vertex is the number of edges adjacent to that vertex. A path between two vertices is a series of distinct, adjacent vertices that connects the vertices. A cycle is a path that starts and ends at the same vertex.

An Example

Example (Finding degrees, paths and cycles)

Find the following using the graph below:

1. The degree of each vertex.
2. The number of paths from vertex 1 to vertex 6.
3. The length of the shortest path from vertex 1 to vertex 6.



An Example
Example (Finding degrees, paths and cycles)
Find the following using the graph below:
1. The degree of each vertex.
2. The number of paths from vertex 1 to vertex 6.
3. The length of the shortest path from vertex 1 to vertex 6.

An Application

Example (Scheduling tournaments)

Suppose six teams are playing in a tournament and each team must play exactly three others. How can this be done? What if only five teams are playing?

An Application
Example (Scheduling tournaments)
Suppose six teams are playing in a tournament and each team must play exactly three others. How can this be done? What if only five teams are playing?

An Impossibility Proof

Example (Not scheduling tournaments)

Prove that it's impossible to schedule a tournament for five teams such that each team plays exactly three others.

An Impossibility Proof
Example (Not scheduling tournaments)
Prove that it's impossible to schedule a tournament for five teams such that each team plays exactly three others.

Look at previous examples and try to find a relationship between degrees and edges.

Another Matching Problem

Example (Utility graph)

Three houses A, B and C must be connected to three utilities E (electric), G (gas) and W (water). Each house must connected to each utility exactly once. What graph represents this situation?

Another Matching Problem
Example (Utility graph)
Three houses A, B and C must be connected to three utilities E (electric), G (gas) and W (water). Each house must connected to each utility exactly once. What graph represents this situation?

Bipartite Graphs

- A bipartite graph is a graph with one group of m vertices and another group of n vertices such that each vertex is adjacent to all of the vertices in the other group and adjacent to none of the vertices in their own group. This graph is denoted by $K_{m,n}$.
- The utility graph is $K_{3,3}$.

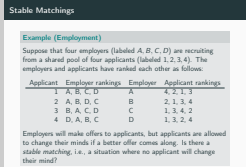
Bipartite Graphs
- A bipartite graph is a graph with one group of m vertices and another group of n vertices such that each vertex is adjacent to all of the vertices in the other group and adjacent to none of the vertices in their own group. This graph is denoted by $K_{m,n}$.
- The utility graph is $K_{3,3}$.

Example (Employment)

Suppose that four employers (labeled A, B, C, D) are recruiting from a shared pool of four applicants (labeled $1, 2, 3, 4$). The employers and applicants have ranked each other as follows:

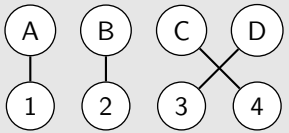
Applicant	Employer rankings	Employer	Applicant rankings
1	A, B, C, D	A	4, 2, 1, 3
2	A, B, D, C	B	2, 1, 3, 4
3	B, A, C, D	C	1, 3, 4, 2
4	D, A, B, C	D	1, 3, 2, 4

Employers will make offers to applicants, but applicants are allowed to change their minds if a better offer comes along. Is there a *stable matching*, i.e., a situation where no applicant will change their mind?



An example of an unstable matching is the following:

Figure 2: A matching that is not stable.



This is because A prefers 2 to 1 and 2 prefers A to B. Therefore, both are inclined to switch.

Stable Matching Algorithm

- The previous problem can be solved by an algorithm. For each round:
  1. Each employer with an open position makes an offer to the best candidate that they haven't made an offer to yet.
  2. Each applicant evaluates their offers and tentatively selects the best offer while rejecting the others.
  3. If there are any open positions remaining, then repeat this process in a new round of offers.

Stable Matching Algorithm
- The previous problem can be solved by an algorithm. For each round:
  1. Each employer with an open position makes an offer to the best candidate that they haven't made an offer to yet.
  2. Each applicant evaluates their offers and tentatively selects the best offer while rejecting the others.
  3. If there are any open positions remaining, then repeat this process in a new round of offers.

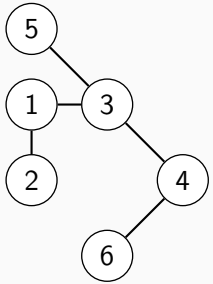
This is known as the Gale-Shapley algorithm. It had previously been used to match residents with hospitals.

Trees

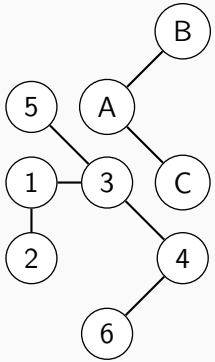
Trees

Definition and Examples

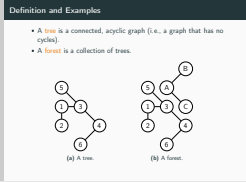
- A tree is a connected, acyclic graph (i.e., a graph that has no cycles).
- A forest is a collection of trees.



(a) A tree.



(b) A forest.



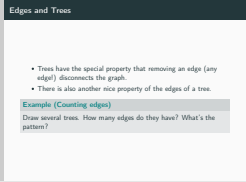
Trees are often useful as *data structures* in computer science. For instance, file directories on a computer can be modeled using trees.

Edges and Trees

- Trees have the special property that removing an edge (any edge!) disconnects the graph.
- There is also another nice property of the edges of a tree.

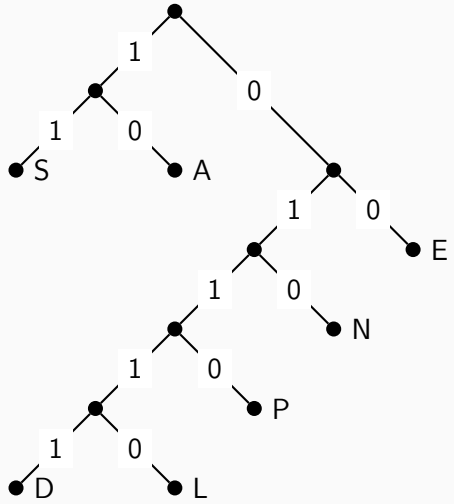
Example (Counting edges)

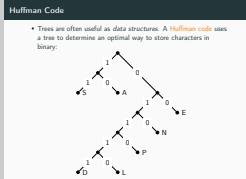
Draw several trees. How many edges do they have? What's the pattern?



Huffman Code

- Trees are often useful as *data structures*. A Huffman code uses a tree to determine an optimal way to store characters in binary:





Huffman codes are constructed using probabilities/frequencies. This particular Huffman code would've been constructed from a set of words where "S" was more common than "L", for instance.

Using Huffman Codes

Example (Encoding and decoding words)

For the previous Huffman code:

1. Encode the words "SAD" and "SPEND".
2. Decode the string "0110011101001011".

Using Huffman Codes
Example (Encoding and decoding words)
For the previous Huffman code:
1. Encode the words "SAD" and "SPEND".
2. Decode the string "0110011101001011".

Constructing Huffman Codes

Example (Code construction)

Construct a Huffman code from the phrase "THIS IS SUPER".

Constructing Huffman Codes
Example (Code construction)
Construct a Huffman code from the phrase "THIS IS SUPER".

Example (Representing a maze as a tree)

Represent the pictured maze as a tree. How can you solve the maze using the tree?

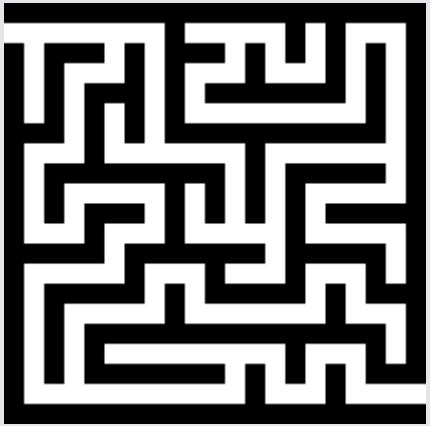


Figure 4: The maze.

Mazes and Trees
Example (Representing a maze as a tree)
Represent the pictured maze as a tree. How can you solve the maze using the tree?
Figure 4: The maze.

Breadth-First Search

- Solving the maze corresponds to finding the shortest path between S (the root) and F (the target) in the tree.
- Such paths can be found using breadth-first search:
  1. Add the root to the queue. While the queue is nonempty, do the following:
  2. Set the current vertex equal to the first vertex in the queue, removing it from the queue.
  3. If the current vertex is the target, then set the current vertex as the parent of the target and erase the queue.
  4. If the current vertex is not the target, then do the following for each neighbor:
     - 4.1 set the parent of the neighbor to the current vertex.
     - 4.2 add the neighbor to the queue.

Breadth-First Search

- Solving the maze corresponds to finding the shortest path between S (the root) and F (the target) in the tree.
- Such paths can be found using breadth-first search:
  1. Add the root to the queue. While the queue is nonempty, do the following:
  2. Set the current vertex equal to the first vertex in the queue, removing it from the queue.
  3. If the current vertex is the target, then set the current vertex as the parent of the target and erase the queue.
  4. If the current vertex is not the target, then do the following for each neighbor:
     - 4.1 set the parent of the neighbor to the current vertex.
     - 4.2 add the neighbor to the queue.

Breadth-First Application

Solving a maze graph

Apply the previous algorithm to find the shortest path from S to F in the following maze graph:

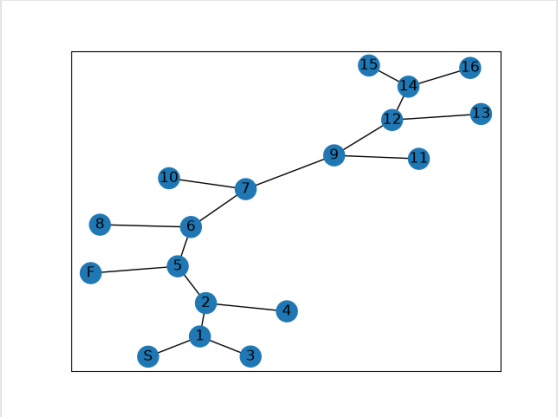


Figure 5: Maze graph.

Breadth-First Application
Solving a maze graph
Apply the previous algorithm to find the shortest path from S to F in the following maze graph:
Figure 5: Maze graph.

BFS algorithm:

1. Add the root to the queue. While the queue is nonempty, do the following:
2. Set the current vertex equal to the first vertex in the queue, removing it from the queue.
3. If the current vertex is the target, then set the current vertex as the parent of the target and erase the queue.
4. If the current vertex is not the target, then do the following for each neighbor:
   - 4.1 set the parent of the neighbor to the current vertex.
   - 4.2 add the neighbor to the queue.

GitHub Repo

Figure 6: GitHub repository for notes.

GitHub Repo

Figure 6: GitHub repository for notes.